AI ASSISTED CODING

Roll no: 2303A52454
B-34

Lab 15 – Backend API Development: Creating RESTful services with AI

Task 1 – Student Attendance API

Task:

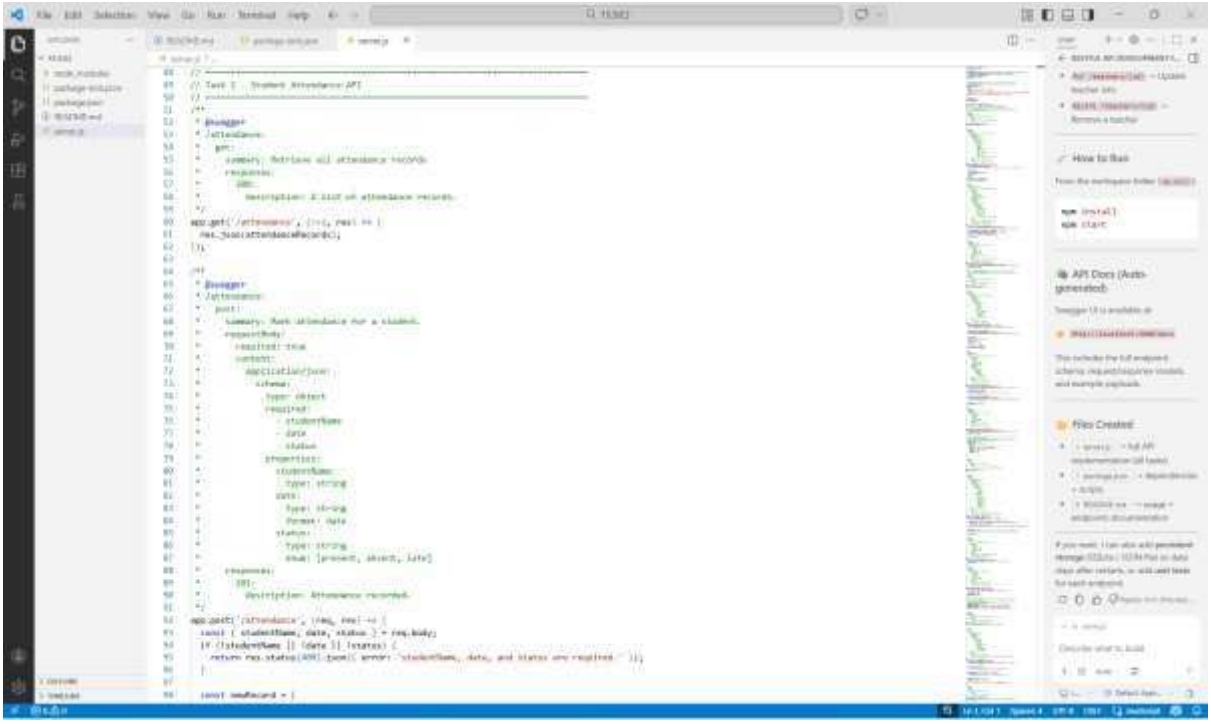
Create a RESTful API for student attendance tracking.

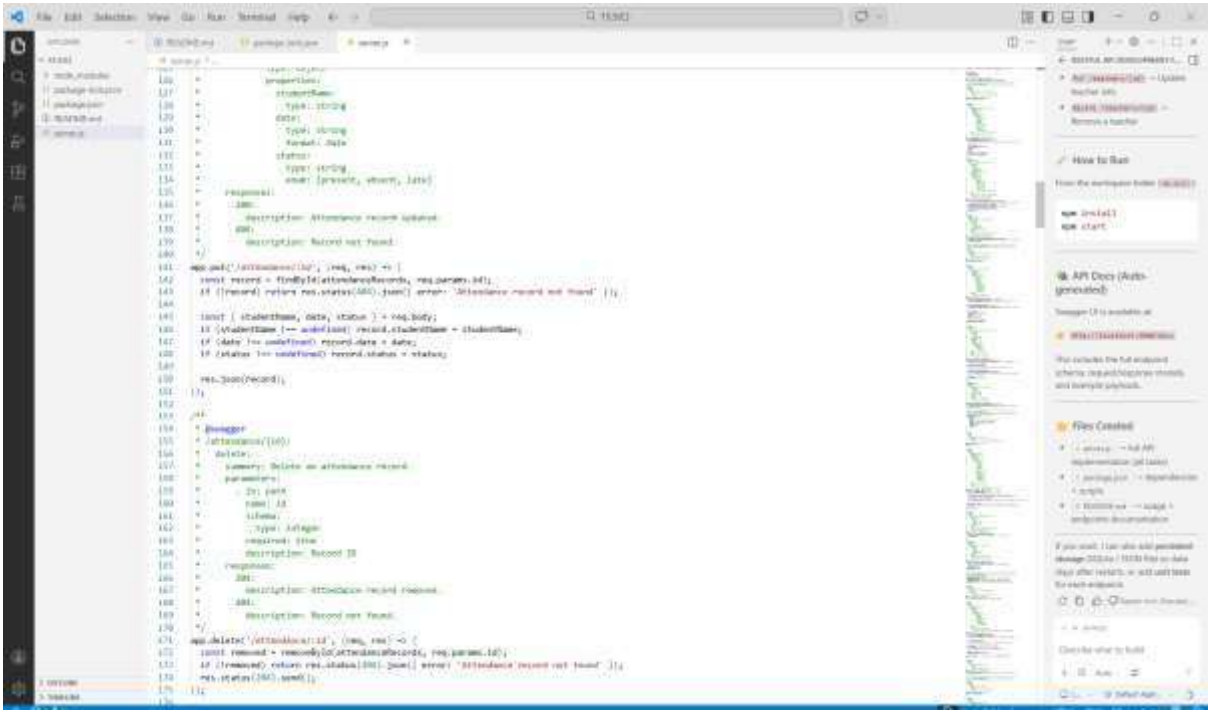
Instructions:

- *Endpoints required: o GET /attendance → View attendance records o POST /attendance → Mark attendance o PUT /attendance/{id} → Update attendance record o DELETE /attendance/{id} → Remove attendance entry*
- *Ensure JSON-based request and response handling.*

Expected Output:

- *API for maintaining and updating attendance records.*





Task 2 – Inventory Management API Task:

Develop a REST API for warehouse inventory management.

Instructions:

• Endpoints required:

o GET /inventory → List all items o POST

/inventory → Add new item o PUT

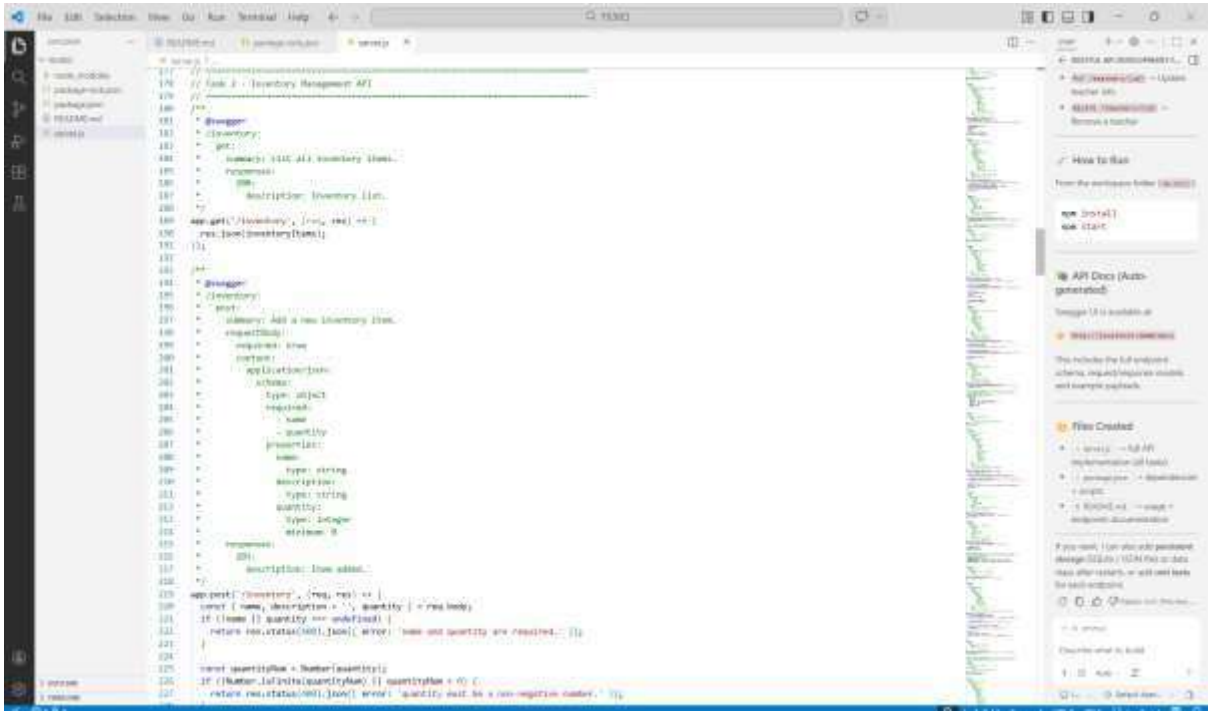
/inventory/{id} → Update stock quantity o

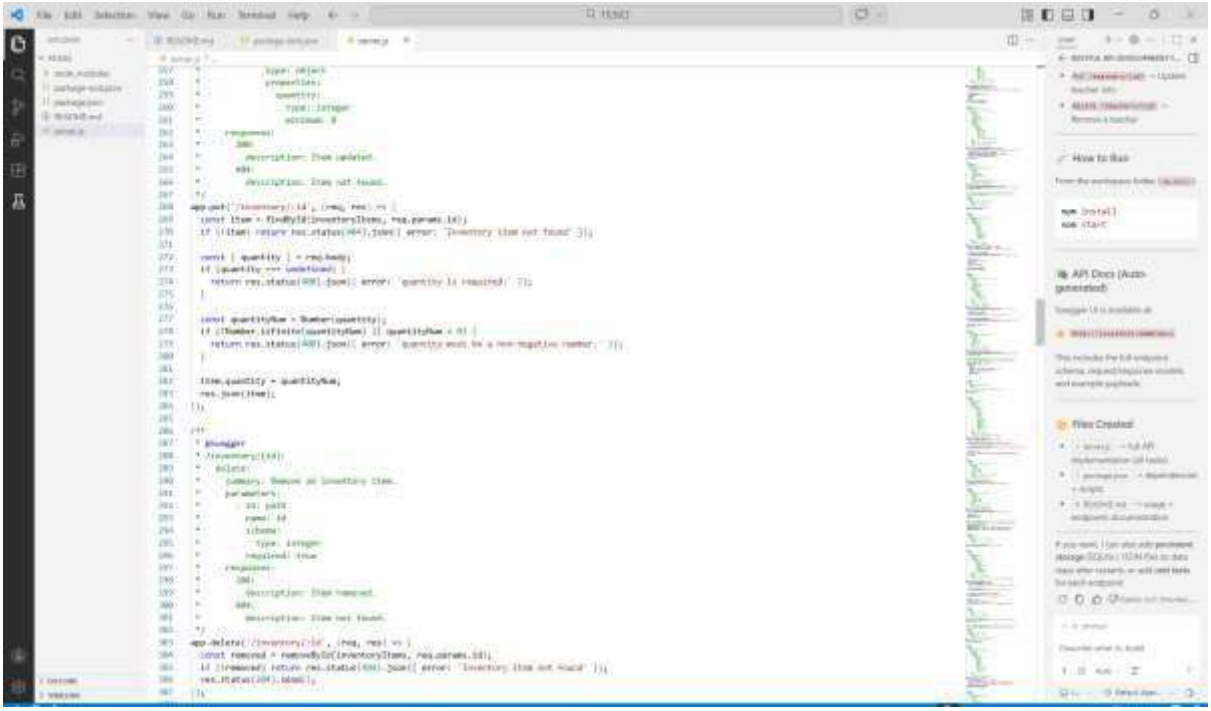
DELETE /inventory/{id} → Remove an item

• Implement validation for stock values.

Expected Output:

• API capable of managing inventory stock levels.





Task 3 – Hotel Room Booking API Task:

Develop a RESTful API for hotel room booking.

Instructions:

• Endpoints required:

o GET /rooms → View available rooms o

POST /rooms/book → Book a room o GET

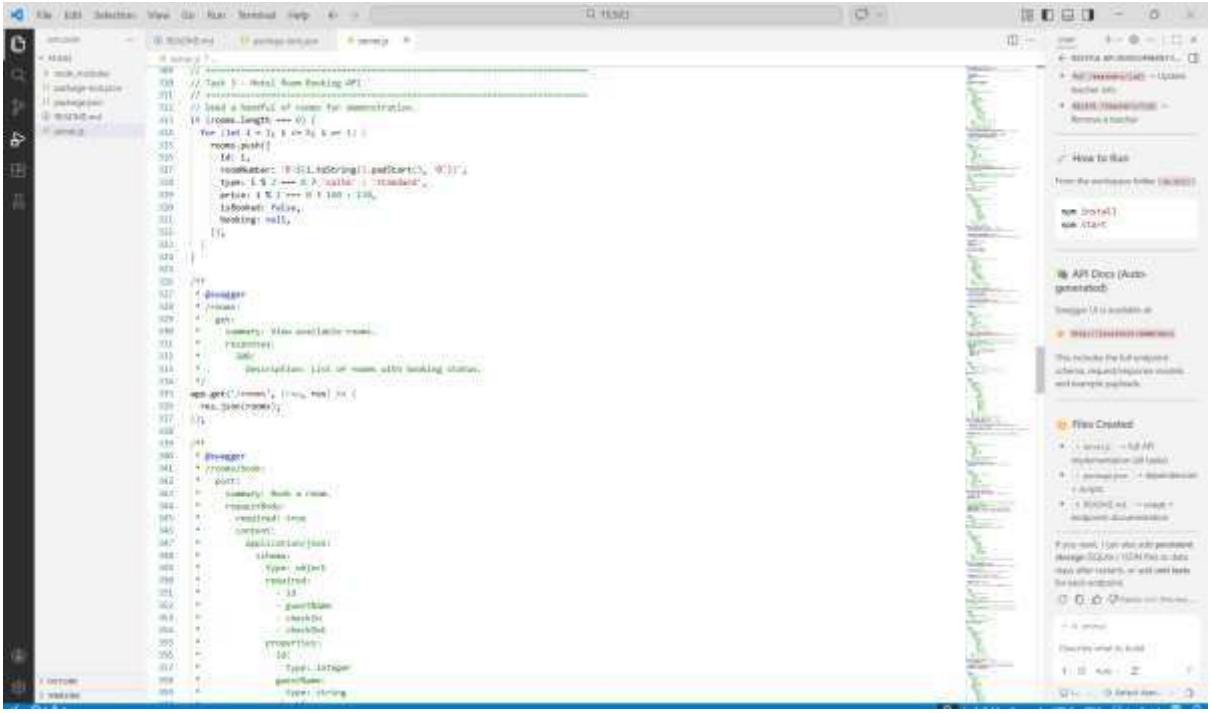
/rooms/{id} → View booking details o

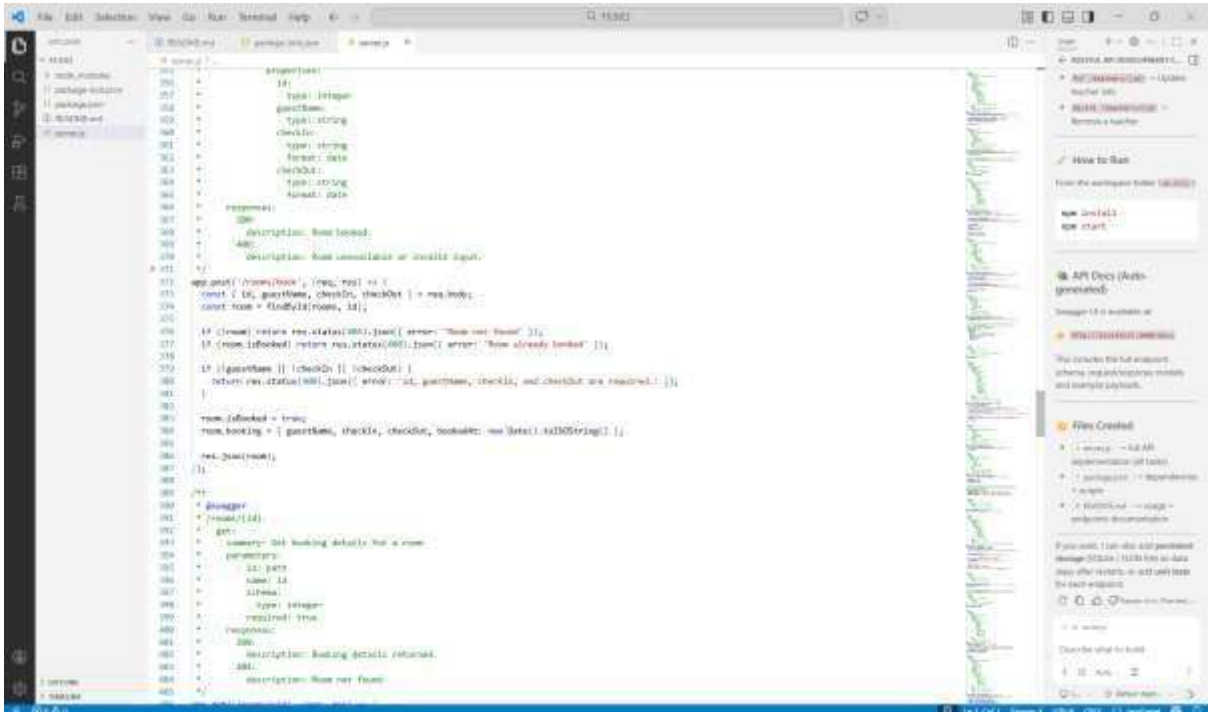
DELETE /rooms/{id} → Cancel booking

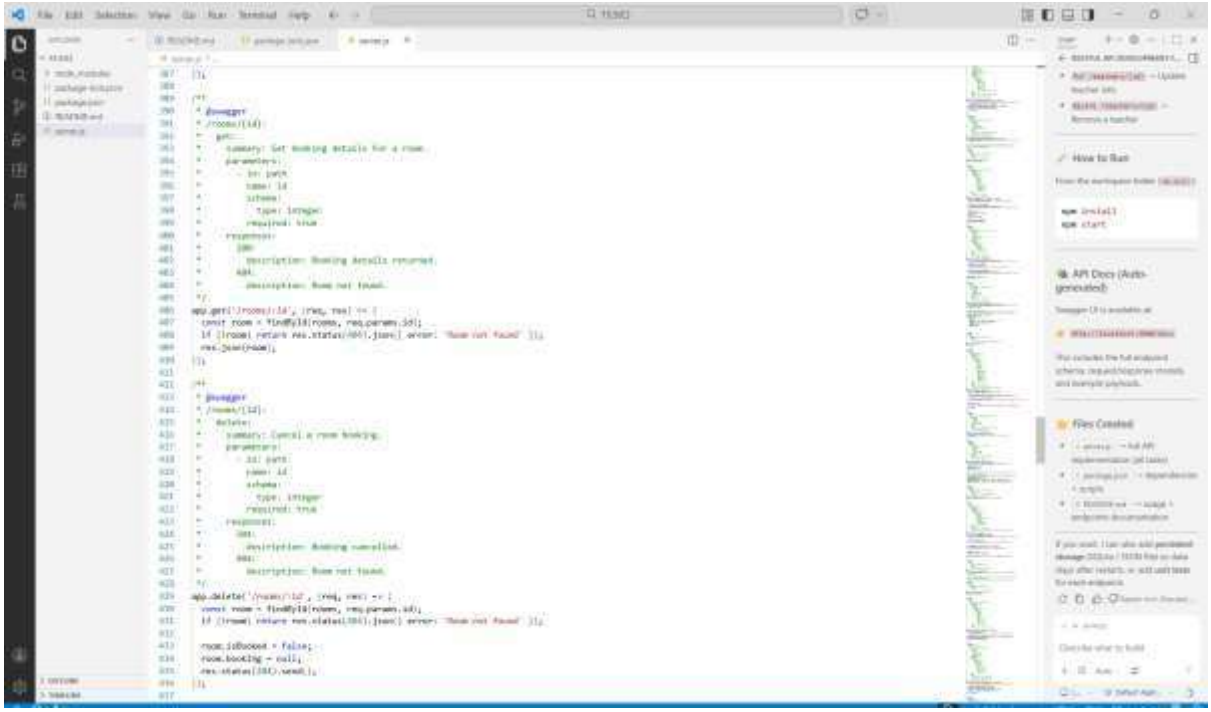
• Implement error handling for unavailable rooms.

Expected Output:

• API simulating hotel room booking operations.







Task 4 – Online Voting API Task:

Create a RESTful API for an online voting system.

Instructions:

• Endpoints required:

o GET /candidates → List candidates

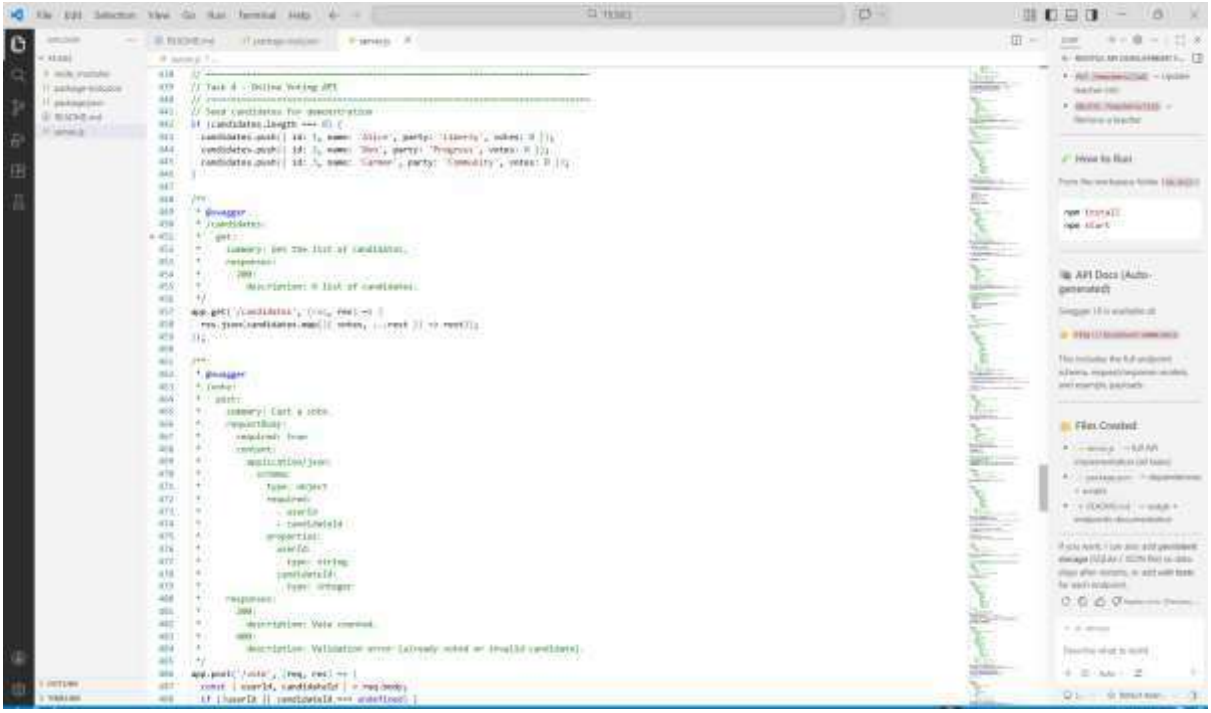
o POST /vote → Cast a vote o GET

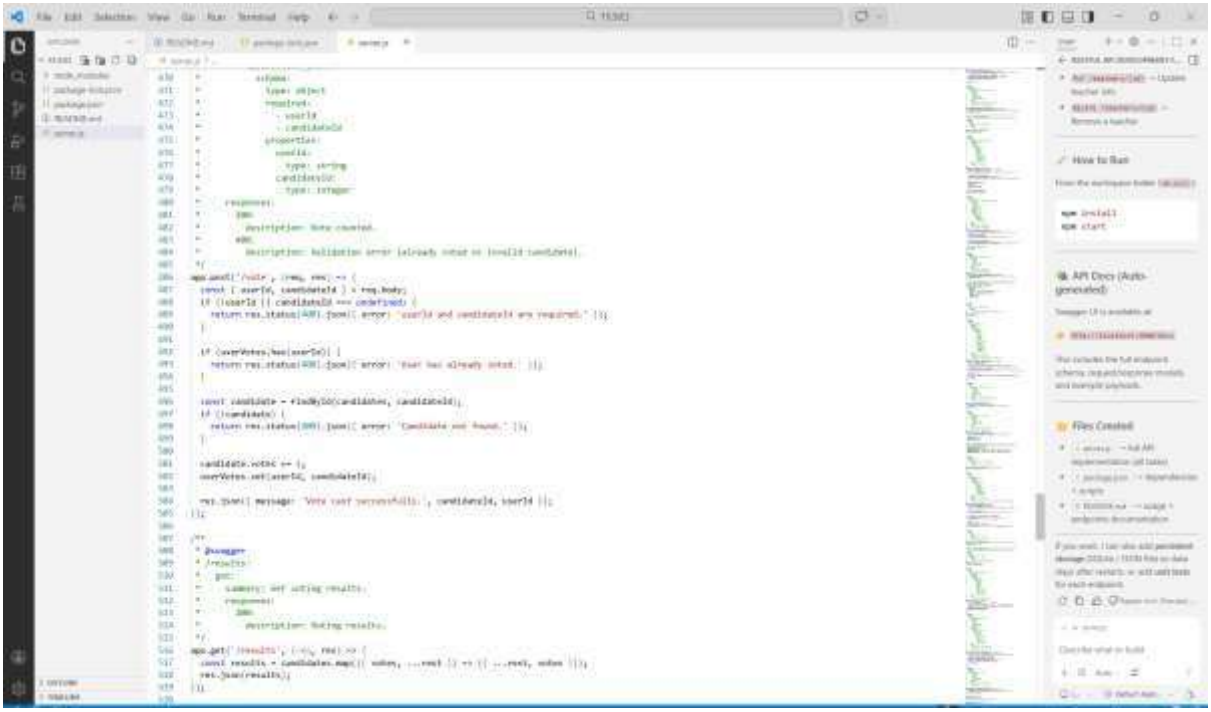
/results → View voting results

• Ensure one vote per user using in-memory validation.

Expected Output:

• API simulating a secure online voting process.





Task 5 – Teacher Management API Task:

Develop a RESTful API to manage teacher records.

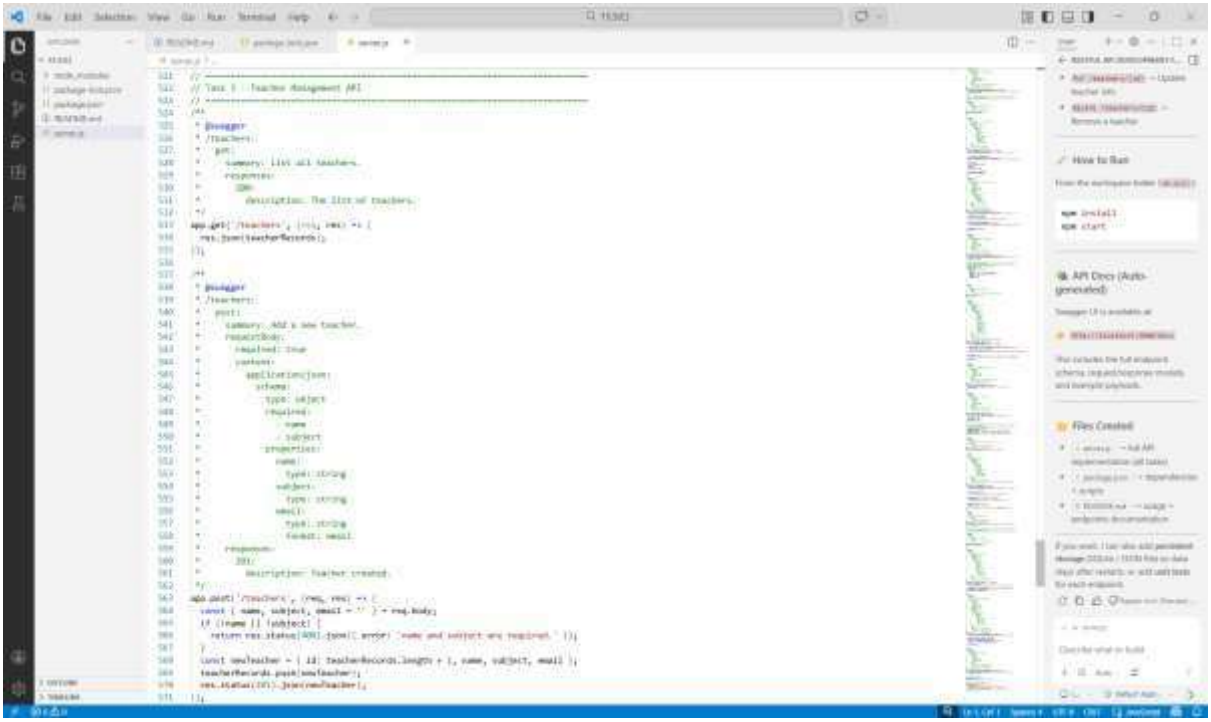
Instructions:

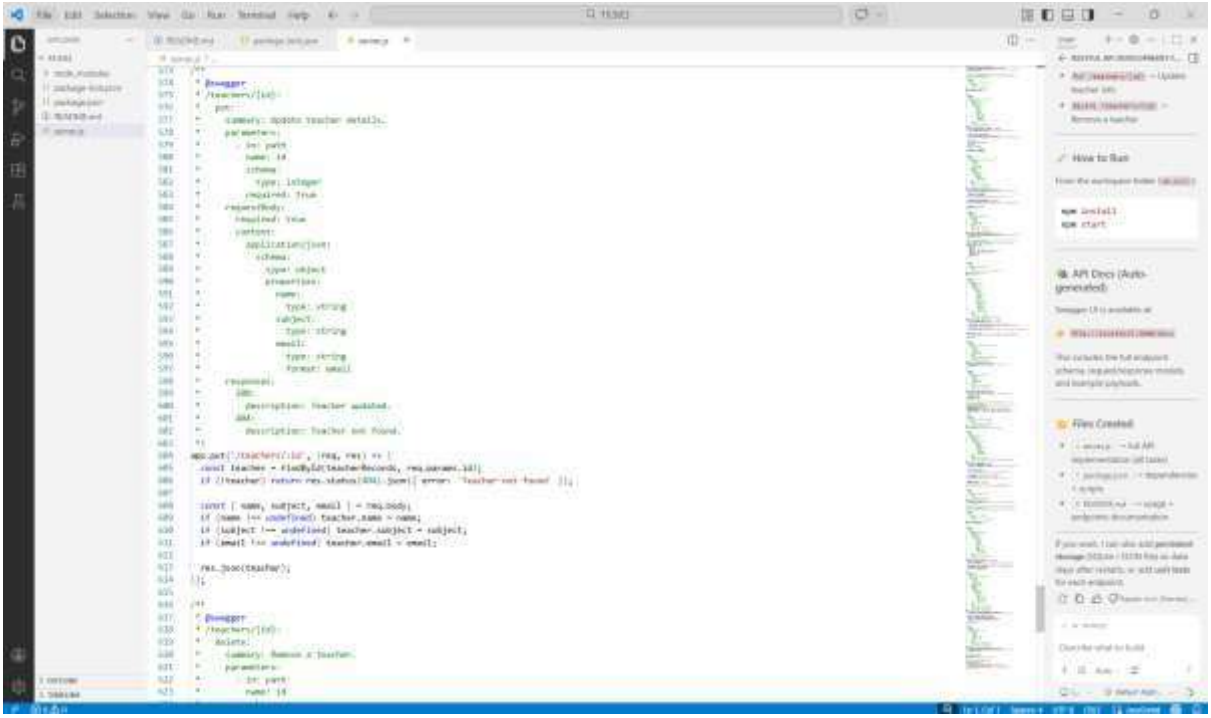
• Endpoints required:

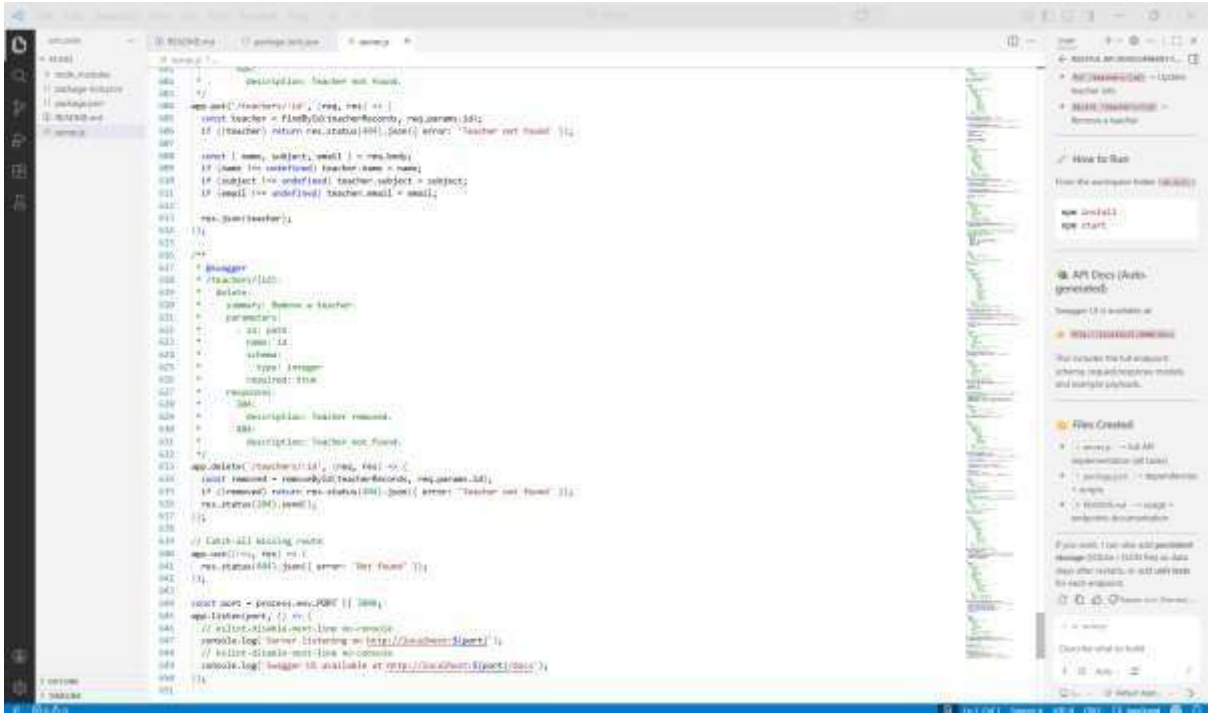
o GET /teachers → List all teachers o POST /teachers → Add a new teacher o PUT /teachers/{id} → Update teacher details o DELETE /teachers/{id} → Remove a teacher • Ensure proper error handling and JSON responses.

Expected Output:

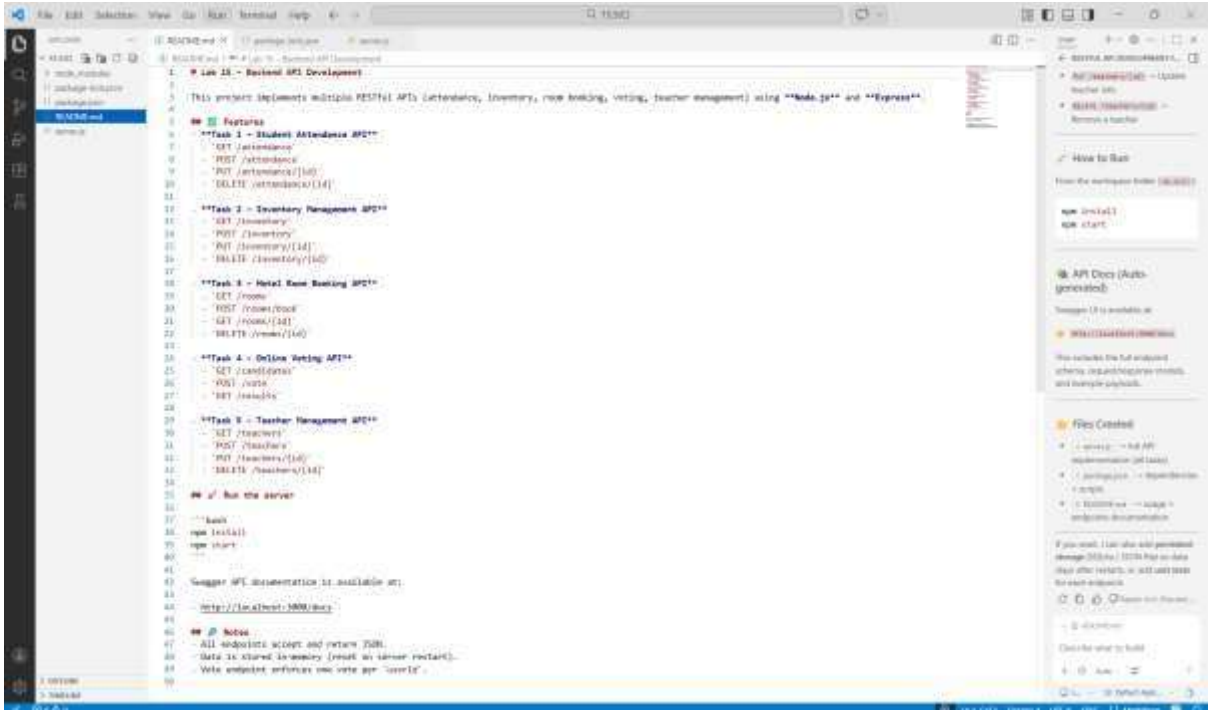
• Functional API for managing teacher information.



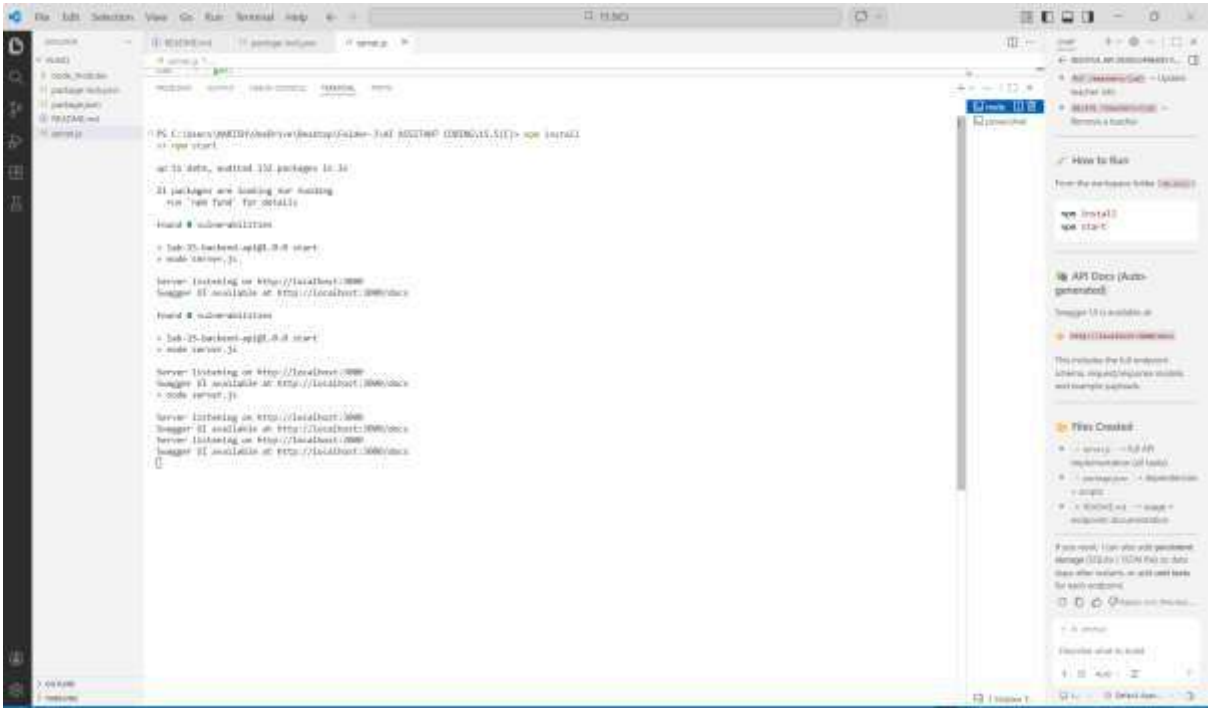




Backend API Development

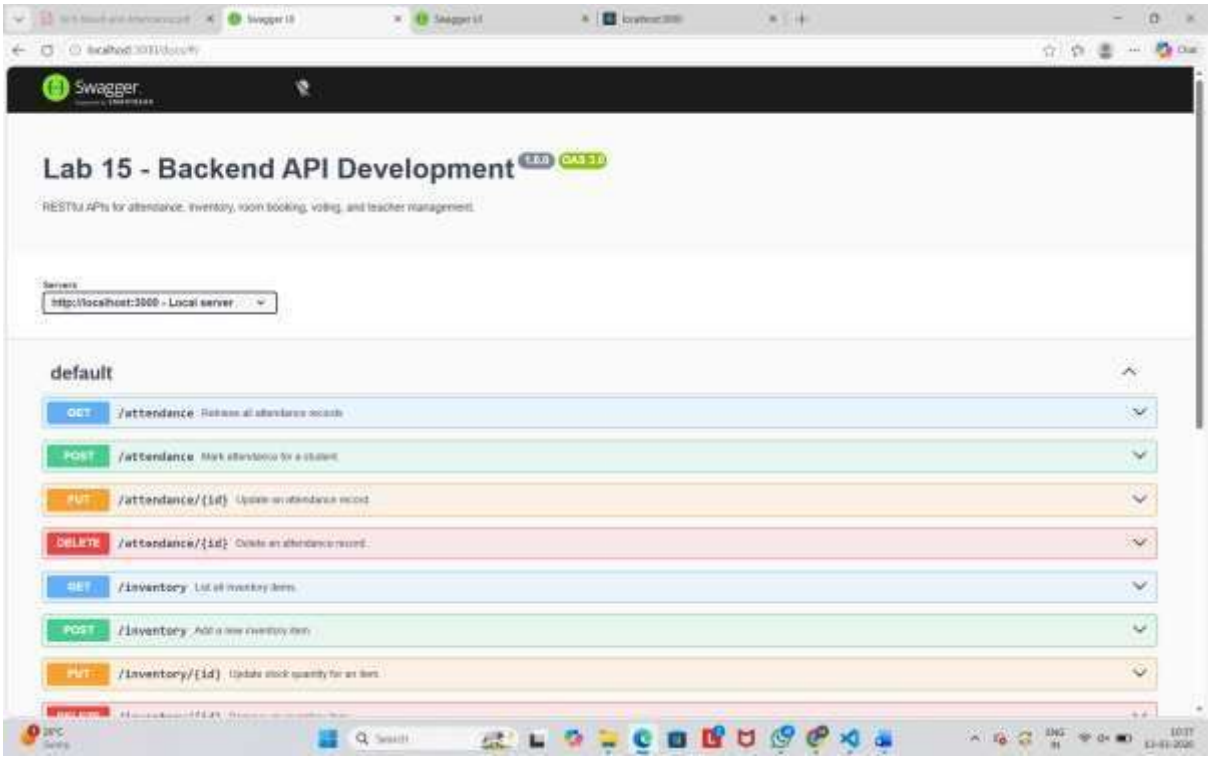


Install Project & Run the Project



Server listening on http://localhost:3000

Swagger UI available at http://localhost:3000/docs http://localhost:3000/docs :


Swagger
Lab 15 - Backend API Development
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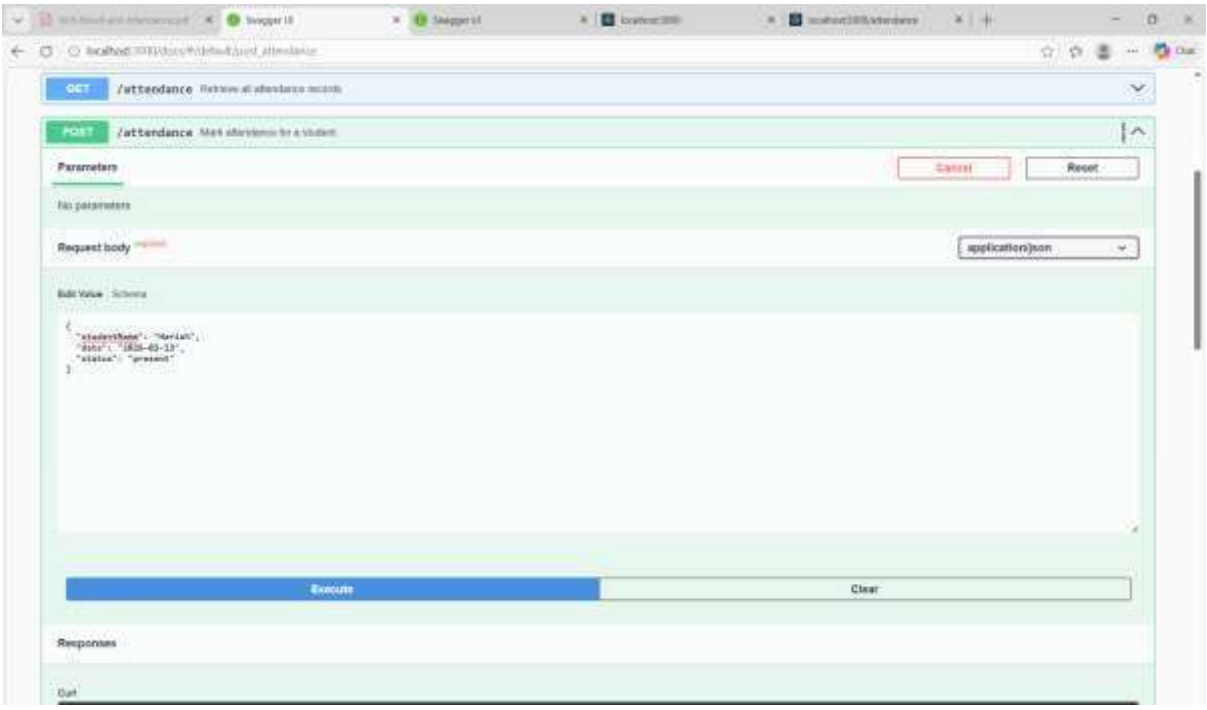
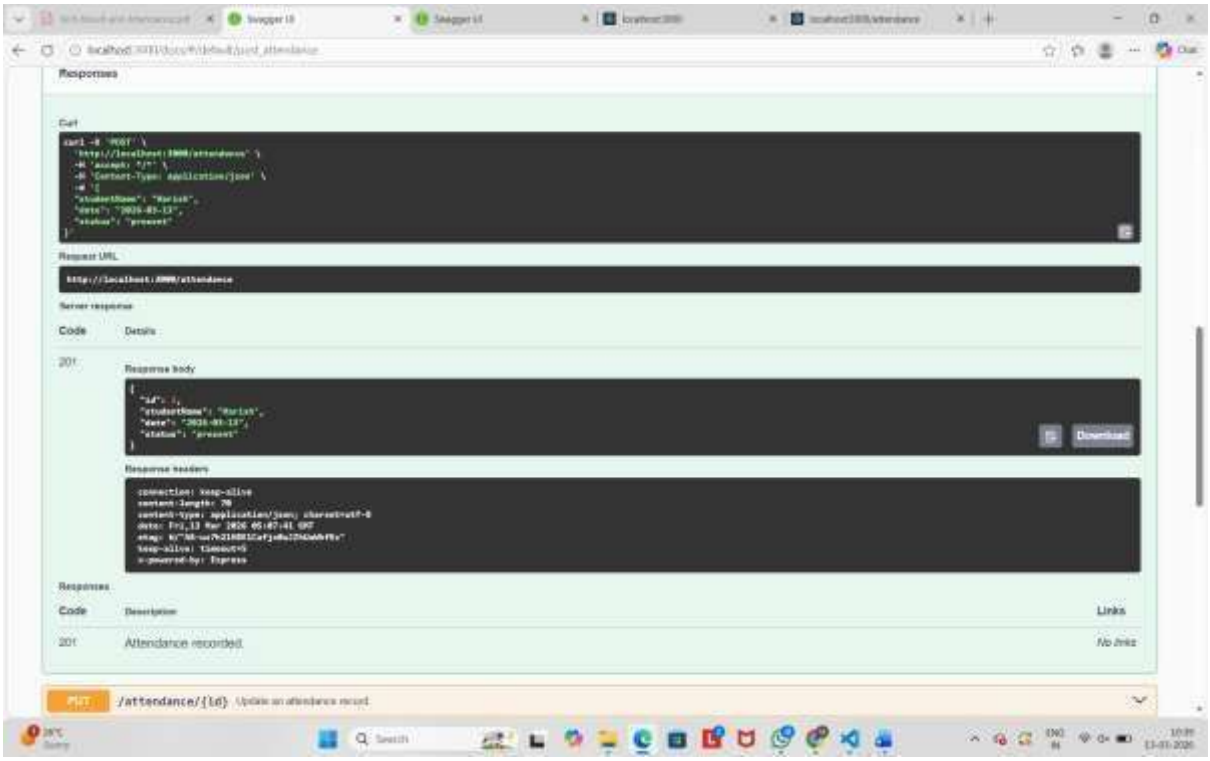
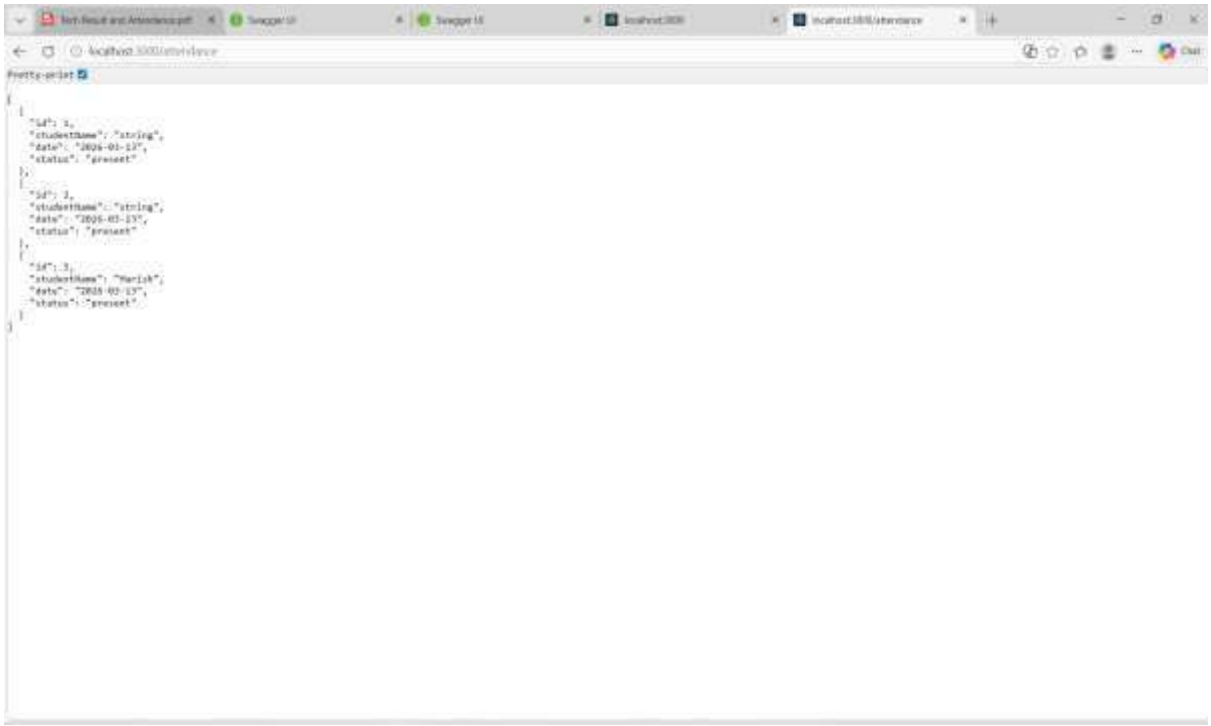
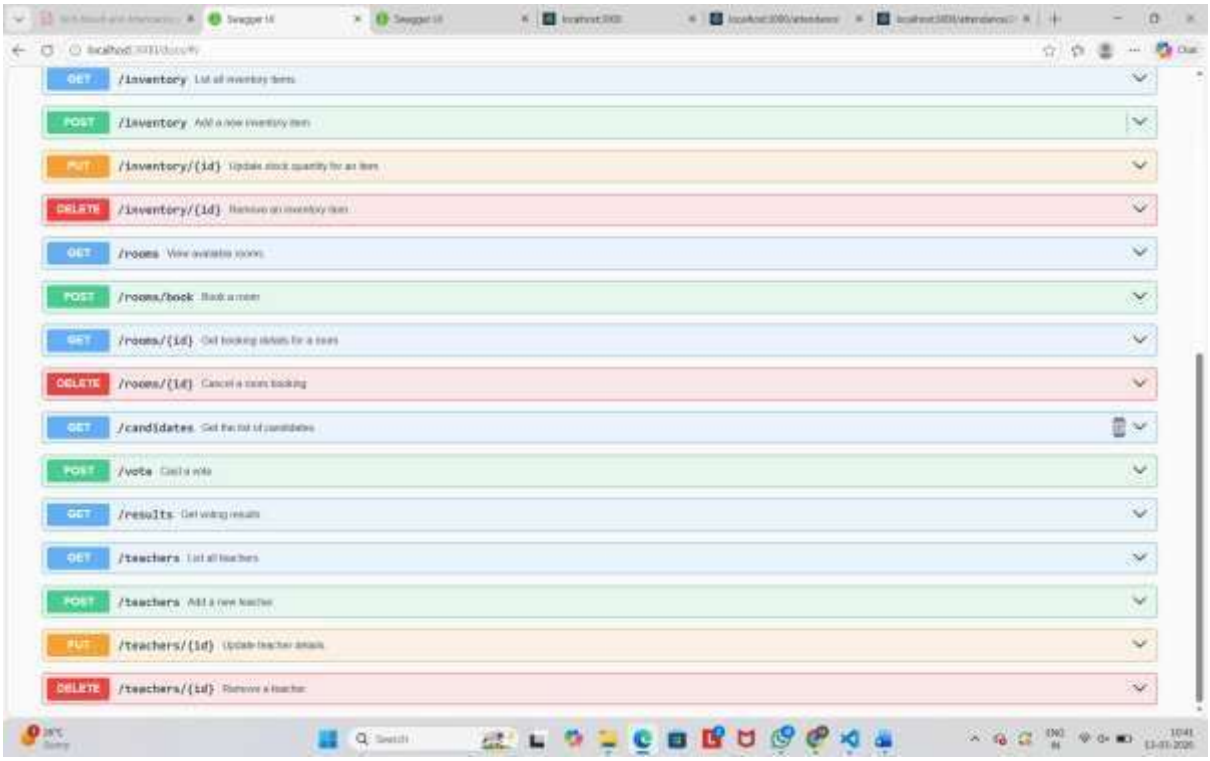
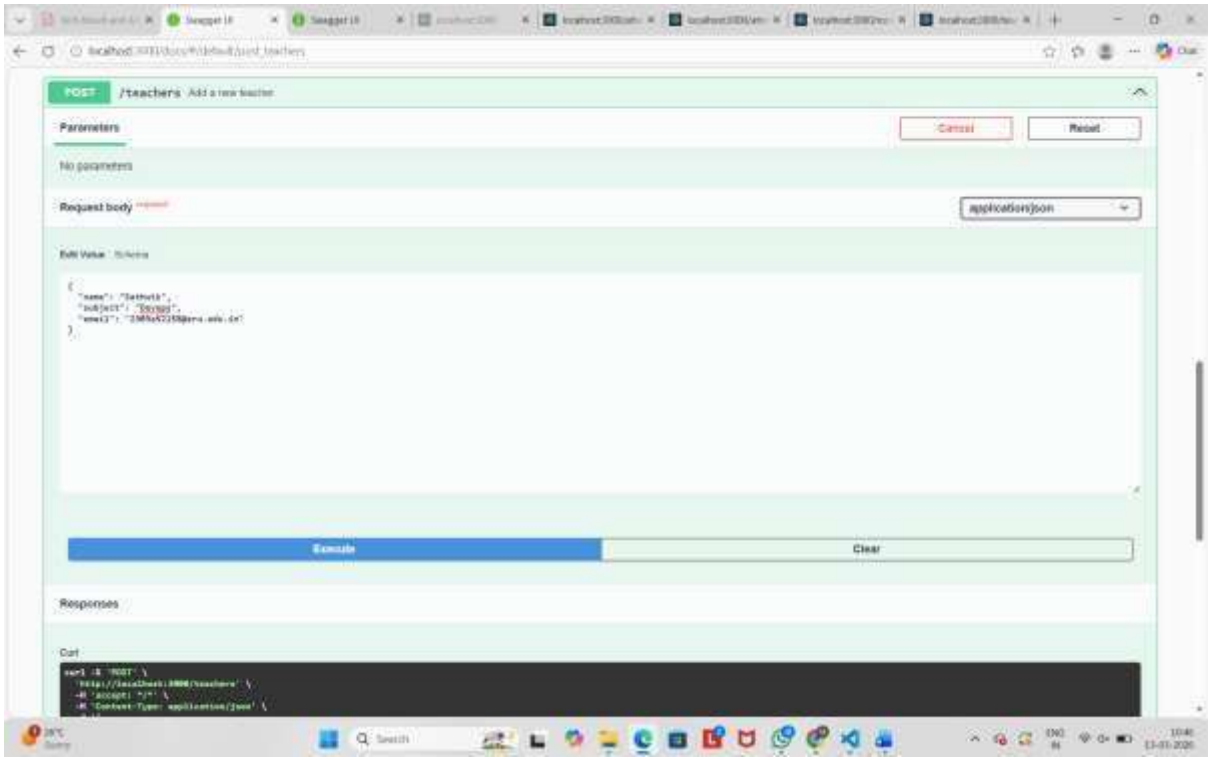
RESTful APIs for attendance, inventory, room booking, voting, and teacher management.
Servers
http://localhost:3000 - Local server
default
GET /attendance Retrieve all attendance records
POST /attendance Mark attendance for a student
PUT /attendance/{id} Update an attendance record
DELETE /attendance/{id} Delete an attendance record
GET /inventory List all inventory items
POST /inventory Add a new inventory item
PUT /inventory/{id} Update stock quantity for an item

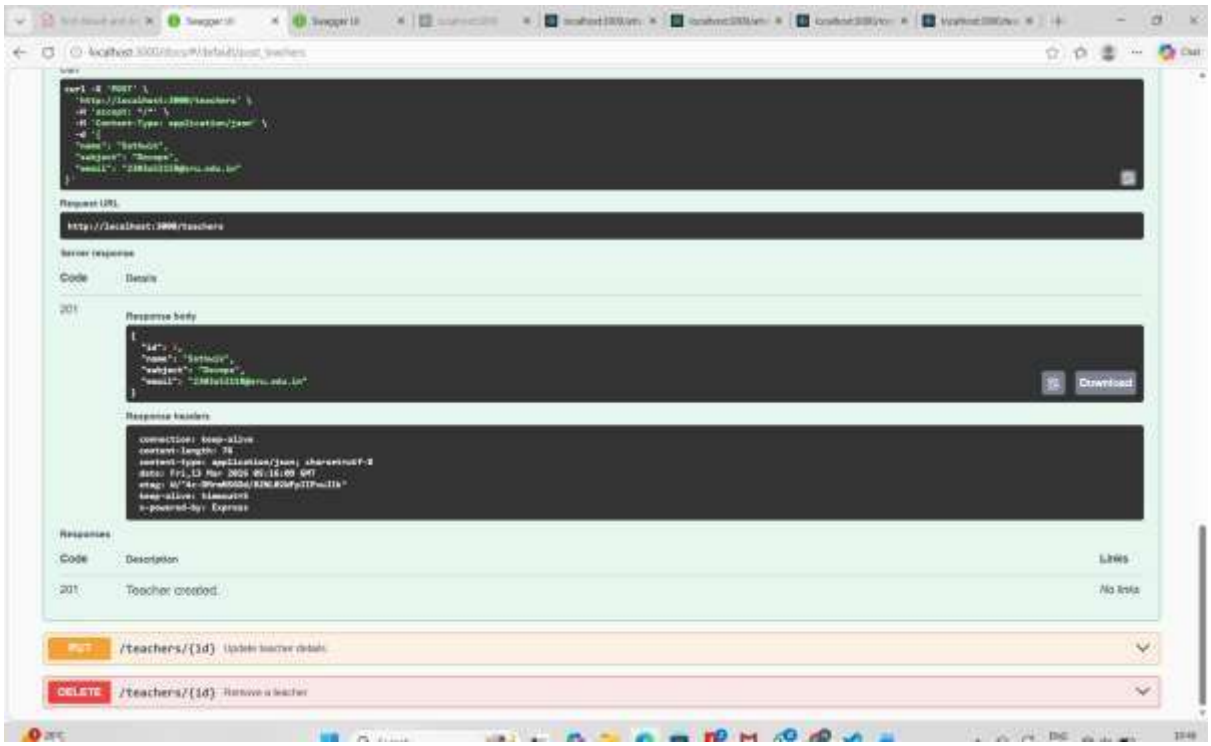

GET /attendance Retrieve all attendance records
POST /attendance Mark attendance for a student
Parameters
Cancel
Reset
No parameters
Request body
application/json
Edit Value
Schema
{
"studentName": "Mariah",
"date": "2026-03-13",
"status": "present"
}
Execute
Clear
Responses
Curl

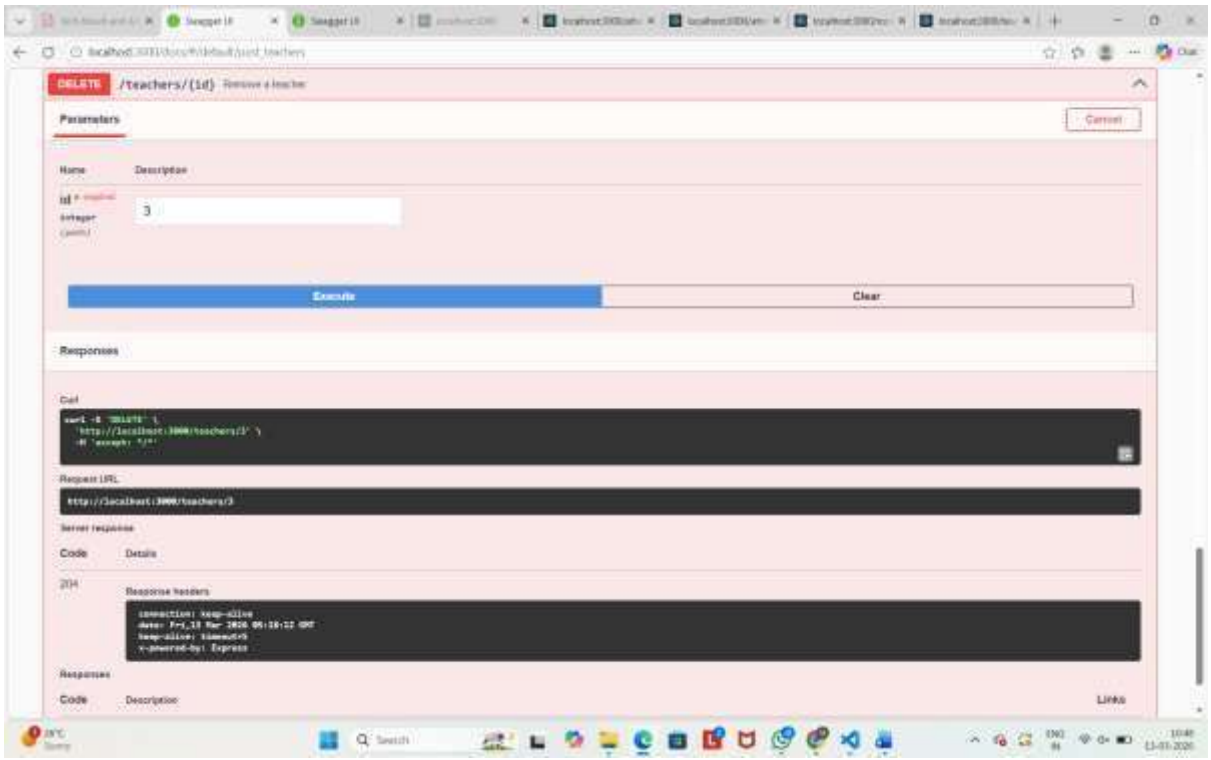

Responses
Curl
Request URL
http://localhost:3000/attendance
Server response
Code
Details
201
Response body
Download
Response headers
Responses
Code
Description
Links
201
Attendance recorded
No links
PUT
/attendance/{id}


Pretty-print
"id": 1,
"studentName": "string",
"date": "2026-03-13",
"status": "present"
"id": 2,
"studentName": "string",
"date": "2026-03-13",
"status": "present"
"id": 3,
"studentName": "Mariah",
"date": "2026-03-13",
"status": "present"


GET /inventory
POST /inventory
PUT /inventory/{id}
DELETE /inventory/{id}
GET /rooms
POST /rooms/book
GET /rooms/{id}
DELETE /rooms/{id}
GET /candidates
POST /vote
GET /results
GET /teachers
POST /teachers
PUT /teachers/{id}
DELETE /teachers/{id}


POST /teachers Add a new teacher
Parameters
Cancel
Reset
No parameters
Request body
application/json
Execute
Clear
Responses
Curl





GITHUB LINK : https://github.com/2303A52387/AI-ASSISTANT-CODING/blob/main/15.zip